

Entropy Rectifying Guidance for Diffusion and Flow Models

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1. Introduction

This report presents a reading summary of two closely related papers in the field of guided image generation with diffusion models. The first and main paper is **Entropy Rectifying Guidance (ERG)** [1] by Berrada et al., published at NeurIPS 2025. The second paper is **Classifier-Free Guidance (CFG)** [2] by Ho and Salimans, which serves as the direct conceptual predecessor of ERG.

Guidance in diffusion models refers to the process of steering the generative sampling toward a desired output, such as matching a text prompt or a class label. While CFG has become the dominant approach in virtually all modern image generation systems, it suffers from well-known trade-offs between image quality, diversity, and prompt consistency. ERG addresses these limitations by proposing a new guidance mechanism that manipulates the attention layers of the model at inference time, without requiring any retraining or additional model.

Section 2 summarizes the ERG paper, covering its contributions, related work, methodology, and experimental results. Section 3 presents the CFG paper, justifies its selection, summarizes its method and results, and discusses its relation to ERG.

2. Summary: Entropy Rectifying Guidance (ERG)

2.1. Motivation and Main Contributions

Diffusion and flow-based generative models have become state-of-the-art tools for image synthesis. A central component of these models is *guidance*: a mechanism that steers the generation toward a desired condition (e.g., a text prompt or a class label). The dominant approach, **Classifier-Free Guidance (CFG)** [2], works by contrasting a conditional prediction with an unconditional one, boosting consistency at inference time. However, as thoroughly analysed by Astolfi et al., CFG introduces inherent trade-offs: increasing guidance strength improves prompt consistency and image quality but *reduces diversity* and causes *color over-saturation* at high scales.

Berrada et al. [1] introduce **Entropy Rectifying Guidance (ERG)** to address these limitations. The key contributions are:

1. **A novel attention-based guidance mechanism** that obtains both a strong and a weak predictive signal from a *single* pre-trained model by manipulating the energy

landscape of its attention layers at inference time, without any retraining.

2. **Universal applicability**: since ERG does not rely on an unconditional forward pass, it applies to unconditional, class-conditional, *and* text-to-image models, a strict generalization over CFG.
3. **Simultaneous improvements** in image quality, diversity, and prompt consistency, breaking the trade-offs observed with CFG. On COCO at 512 resolution, ERG achieves +22 points in Density and +3.8 in VQAScore over CFG with identical computational cost (NFE = 2).

2.2. Key Related Works

2.2.1. Guidance methods

Classifier Guidance [3] first demonstrated high-quality guided sampling by mixing score estimates with gradients from a noise-robust auxiliary classifier. **CFG** [2] eliminates the classifier by jointly training conditional and unconditional models; it remains the de-facto standard. **AutoGuidance** replaces the unconditional term with a weaker version of the same model (an earlier checkpoint or smaller architecture), requiring two model variants. **SEG** uses Gaussian blurring of attention queries to weaken predictions for U-Net architectures (3 forward passes for conditional sampling). **CADS** adds annealed Gaussian noise to conditioning tokens to increase diversity. **APG** resolves CFG over-saturation via orthogonal projection of the guidance signal. ERG can be combined with CADS and APG.

2.2.2. Hopfield energy formulation

ERG builds on the theoretical connection between self-attention and Modern Hopfield Networks [4]. A standard attention layer is equivalent to one gradient descent step minimizing the energy:

$$E(\xi; X) = \frac{1}{2}\xi^\top \xi - \text{LSE}(X^\top \xi, \beta)$$

where ξ are queries, X are keys, $\beta = 1/\sqrt{d}$, and $\text{LSE}(x, \beta) = \frac{1}{\beta} \log \sum_i \exp(\beta x_i)$. This principled formulation enables ERG to modify the energy landscape to produce weaker predictions suitable for guidance contrast.

2.3. Methodology

2.3.1. Manipulating the Hopfield energy

ERG modifies the Hopfield energy by introducing two hyperparameters α and τ :

$$E(\xi; X) = \frac{1}{2}\xi^\top \xi - \alpha \cdot \text{LSE}(X^\top \xi, \tau\beta) \quad (1)$$

Here $\tau > 0$ rescales the softmax temperature (low τ spreads attention uniformly, high τ sharpens it), and α modulates the pattern-matching weight. Taking K gradient descent steps with step size λ yields the update rule:

$$Q \leftarrow Q - \lambda(Q - \alpha \cdot \text{softmax}(\tau \cdot QK^\top)V) \quad (2)$$

Setting $\alpha = \lambda = \tau = K = 1$ recovers standard attention exactly; other values produce a *weaker* attention operation.

2.3.2. Two-pronged application: C-ERG and I-ERG

C-ERG (Condition-ERG): A temperature τ_c is introduced into every self-attention layer of the text encoder (Llama3-8B or Flan-T5-XL). This produces a weakened conditional embedding c^{τ_c} that serves as the negative conditioning signal.

I-ERG (Image-ERG): Parameters $\{\alpha, \lambda, \tau_i, K\}$ modify selected attention layers in the image denoising transformer. I-ERG is applied only after a kickoff timestep threshold ϕ (to preserve early global structure) and on specific intermediate layers identified by ablation.

2.3.3. Final ERG update

$$\Psi_{\text{ERG}}(x, c, t) = w \cdot D(x, c, t) + (1 - w) \cdot D_\xi(x, c^{\tau_c}, t; \Theta_\xi) \quad (3)$$

where D is the standard denoiser, D_ξ uses entropy-rectified attention (I-ERG), and $\Theta_\xi = \{\alpha, \lambda, \tau_i, K\}$. Unlike CFG, both terms use *conditional* embeddings; the difference is the quality of internal computation.

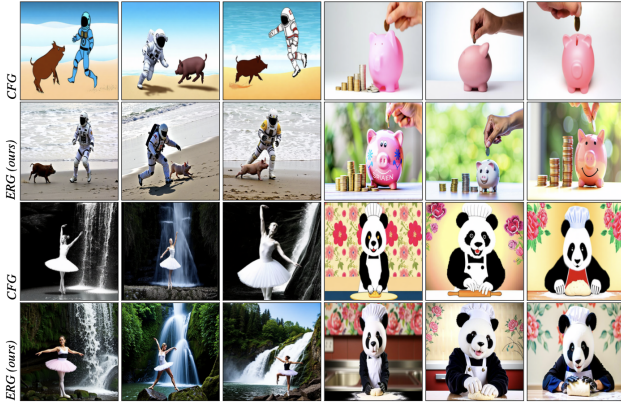


Figure 1: **Qualitative comparison: CFG (rows 1 & 3) vs. ERG (rows 2 & 4).** Same seeds across columns. ERG generates more diverse and visually detailed outputs, e.g., richer backgrounds and more varied compositions.

2.4. Experimental Setup

Architectures. Two main models are evaluated: (1) a class-conditional DiT XL/2 (790M params) on ImageNet at 256 and 512; (2) a 1.9B MMDiT-like text-to-image model using Llama3-8B and Flan-T5-XL as text encoders, trained on 320M proprietary pairs + YFCC100M.

Evaluation protocol. All models sampled with 50 Euler steps. Quality: FID and Density; diversity: Coverage;

consistency: CLIPScore and VQAScore. Reference: 40k COCO’14 validation pairs for text-to-image; ImageNet validation for class-conditional.

2.5. Main Results

2.5.1. Text-to-image

Table 1 compares ERG against CFG and recent baselines on COCO’14. ERG alone achieves the best Density (+22.0 over CFG), best CLIPScore and VQAScore, at NFE = 2. ERG+APG achieves best Coverage (80.50) and FID (11.37), and Pareto-dominates APG across all three metrics simultaneously (Fig. 2).

Table 1: **Text-to-image on COCO’14.** Best single-method values **bolded**. * = adapted from U-Net to transformer.

Method	FID↓	Den.↑	Cov.↑	CLIP↑	VQA↑	NFE
CFG	12.81	98.24	71.12	26.45	70.15	2
APG	11.88	104.07	73.06	26.54	72.47	2
CADS	11.93	101.01	72.99	26.76	73.36	2
PAG*	12.75	107.21	72.20	26.80	73.32	2
SAG*	11.68	103.58	72.74	26.81	72.16	2
SEG*	16.87	87.77	61.91	26.86	73.59	3
AutoGuid.	16.62	87.02	62.75	26.59	73.53	2
ERG	13.62	120.25	73.21	26.86	73.96	2
ERG+APG	11.37	115.08	80.50	26.74	73.55	2
ERG+CADS	12.87	128.54	76.23	26.75	73.45	2

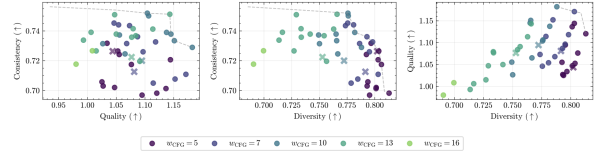


Figure 2: **Pareto fronts: ERG+APG (dots) vs. APG (crosses).** Each color is a guidance scale w . ERG+APG Pareto-dominates APG across all pairs of metrics (quality, diversity, consistency).

2.5.2. Unconditional and class-conditional generation

Table 2 shows unconditional results (empty prompt). ERG improves FID from 101.50 to 36.25 and Coverage from 3.63 to 51.59, far exceeding all baselines. For class-conditional ImageNet, ERG achieves best Density at 256 (141.96) and 512 (163.63), matching CFG on FID at 256 (3.67).

Table 2: **Unconditional generation.** ERG surpasses all baselines significantly, especially in Coverage (diversity).

Method	FID↓	Density↑	Coverage↑
No guidance	101.50	8.99	3.63
SAG	39.25	46.22	30.70
PAG*	41.50	45.65	30.51
SEG*	37.75	55.56	34.79
AutoGuid.	39.50	48.26	34.71
ERG	36.25	55.84	51.59



Figure 3: **Unconditional generation: no guidance (top) vs. ERG (bottom).** ERG produces coherent, realistic images even without any conditioning signal.

2.6. Ablation Study

Table 3 decomposes ERG into C-ERG and I-ERG. C-ERG (τ_c) is the primary driver of consistency (CLIPScore: +0.28, VQAScore: +2.95), while I-ERG (τ_i) mainly drives quality and diversity (Density: +10.73). Setting $K = 1, \lambda = 1$ is the recommended default as multi-step updates give marginal gains.

Table 3: **Ablation: cumulative ERG components.** ✓=active, ×=off. Each component adds consistent improvements.

τ_c	τ_i	λ	FID↓	Den.↑	Cov.↑	VQA↑
×	×	×	12.81	98.24	71.12	70.15
✓	×	×	13.06	109.52	72.06	73.10
✓	✓	×	13.62	120.25	73.21	73.96
✓	✓	✓	13.62	123.65	74.07	74.67

3. Second Paper: Classifier-Free Diffusion Guidance

3.1. Choice and Justification

We select **Classifier-Free Guidance (CFG)** [2] by Ho and Salimans as the companion paper. CFG is the *direct conceptual predecessor* of ERG: it defines the structural form (linear combination of strong and weak predictions) that ERG generalizes, and it is the primary baseline against which ERG demonstrates all its improvements. Understanding CFG’s limitations (diversity collapse, over-saturation, required unconditional training) is essential to appreciating what ERG contributes.

3.2. Summary of CFG

3.2.1. Problem setting

Prior to CFG, high-quality conditional sampling required *classifier guidance* [3]: adding gradients from a noise-robust classifier $p_\phi(c|z_\lambda)$ at each denoising step. This requires a separately trained auxiliary model and raises questions about whether improvements arise from genuine conditioning or adversarial exploitation of the classifier.

3.2.2. Method

CFG trains a single network to perform both conditional and unconditional denoising. The conditioning label c is randomly nulled to $\emptyset$ with probability p_{uncond} during training (typically 0.1–0.2). At inference, the guided score estimate is:

$$\tilde{\epsilon}_\theta(z_\lambda, c) = (1 + w) \epsilon_\theta(z_\lambda, c) - w \epsilon_\theta(z_\lambda) \quad (4)$$

The guidance weight $w \geq 0$ controls the quality-diversity trade-off: $w = 0$ gives unguided diversity; high w increases sample fidelity at the cost of coverage. The formulation is inspired by an implicit classifier $p^i(c|z_\lambda) \propto p(z_\lambda|c)/p(z_\lambda)$, though in practice the connection is approximate since ϵ_θ is an unconstrained neural network.

3.2.3. Results

CFG achieves state-of-the-art IS/FID trade-offs on class-conditional ImageNet at 64×64 and 128×128 without any external classifier. At $w = 0.3$, FID on 128×128 outperforms classifier-guided ADM-G [3]. High guidance scales (e.g., $w = 4$) yield very high IS (421 vs 82 unguided) at significantly degraded FID, illustrating the quality-diversity trade-off inherent to CFG.

3.3. Relation Between CFG and ERG

3.3.1. Shared structural form

Both methods follow the same guidance template: combine a high-quality positive prediction with a lower-quality negative prediction, and extrapolate away from the latter. In CFG (Eq. 4): $(1 + w) \cdot p_{\text{cond}} - w \cdot p_{\text{uncond}}$; in ERG (Eq. 3): $w \cdot D_{\text{normal}} + (1 - w) \cdot D_{\text{weakened}}$.

3.3.2. What ERG overcomes

CFG has three fundamental constraints that ERG directly addresses:

(1) **Unconditional training requirement.** CFG dedicates training steps to unconditional generation solely to produce the negative signal, wasting model capacity. ERG produces the negative signal purely at *inference time* by degrading attention quality, requiring zero training changes and working on any pre-trained model.

(2) **Quality-diversity trade-off.** CFG’s unconditional prediction pulls all samples toward mode-heavy regions, reducing diversity when guidance is increased. ERG’s negative signal is a “same-condition, lower-quality” prediction, a targeted contrast that can simultaneously improve all three generation facets (Table 1).

(3) **Inapplicability to unconditional sampling.** CFG cannot be applied when unconditional generation is the goal. ERG applies to any attention-based model regardless of conditioning mode, as demonstrated in Table 2.

3.3.3. Nature of the negative signal

This is the deepest conceptual difference. CFG’s negative signal changes the *input*: removing conditioning produces a very different prediction that steers away from specific

content. ERG’s negative signal degrades the *internal computation*: attention quality is reduced while keeping the conditioning intact. This “same-input, degraded-process” contrast avoids conflating guidance direction with absence of conditioning information, which is why ERG can improve consistency simultaneously with quality and diversity, something CFG fundamentally cannot achieve.

3.3.4. Computational parity

Both CFG and ERG require exactly two forward passes per denoising step ($\text{NFE} = 2$), as shown in Table 1. ERG’s attention modifications add negligible overhead within existing forward passes. By contrast, SEG requires three passes and AutoGuidance requires loading two checkpoints. ERG is therefore a practical drop-in improvement over CFG at identical cost.

In summary, CFG provides the algorithmic skeleton—contrast two predictions and scale the difference—that ERG inherits and extends. ERG’s contribution is to show that the *nature* of this contrast, not merely its existence, determines whether guidance improves or degrades the three axes of quality, diversity, and consistency.

4. Conclusion

This report reviewed two foundational papers in guided diffusion model sampling. CFG [2] established the standard approach of contrasting conditional and unconditional predictions to improve sample quality, but at the cost of reduced diversity and a mandatory unconditional training phase. ERG [1] builds directly on this framework and addresses its core limitations by replacing the unconditional negative signal with one derived from entropy-rectified attention layers, requiring no additional training or model.

The key takeaway is that the *nature* of the negative guidance signal matters as much as its existence. By obtaining a weaker signal through internal attention degradation rather than input removal, ERG simultaneously improves quality, diversity, and consistency, three objectives that CFG can only trade off against each other. ERG also uniquely extends guidance to purely unconditional generation, a setting where CFG is by definition inapplicable. Together, these two papers illustrate an important design principle: inference-time modifications to model internals can be a more powerful and flexible tool than training-time modifications to model inputs.

References

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